



Strengthening Customer Data Accuracy in the Banking Sector with ML-Based Deduplication

Case Study



Estimated Annual Business Impact

\$ 1M+

Saved in compliance
handling, identity resolution,
and fraud processing.

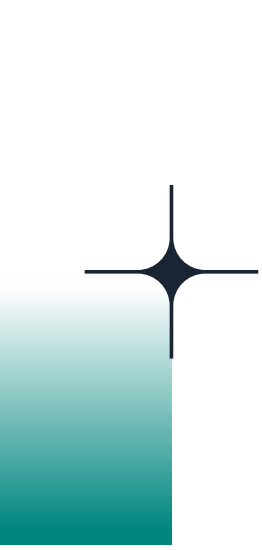
3X

Faster customer onboarding
through real-time data
match and merge.

75%

Reduction in effort with ML
driven automation





Business Problem

In banking, accurate customer identity is critical to compliance (KYC/AML), fraud prevention, and delivering personalized financial products. However, banks often struggle with fragmented or duplicate records due to channel silos, legacy systems, and inconsistent onboarding practices:



10–14% of customer records are duplicated across branches, online apps, credit card systems, and legacy cores.



Rule-based matching fails to capture customer variations such as abbreviated names, address mismatches, or joint-account inconsistencies.



Onboarding and service agents spend hours resolving identity discrepancies across systems.



Regulatory compliance (e.g., AML screening) and customer trust are undermined by inconsistent data.

High-Level Solution

ML-BASED IDENTITY RESOLUTION

- Applied similarity scoring to names, addresses, phone numbers, SSNs, and account metadata across systems.
- Trained ML classifiers on historical match/non-match patterns to detect duplicates beyond exact rules.

ACTIVE LEARNING WITH AUDIT TRAIL

- Used human-in-the-loop for ambiguous cases to ensure compliance-grade accuracy.
- Maintained an auditable log of training examples and resolution actions.

GOLDEN CUSTOMER RECORD CREATION

Merged validated duplicates into a single customer profile for KYC, AML, and CX purposes.

TECHNOLOGY STACK

- Python + Dedupe: Deduplication and record linkage.
- AWS Glue + Redshift: Scalable data ingestion and warehouse.
- Apache Airflow: Job orchestration and scheduling.
- Streamlit UI: Compliance-friendly validation interface.
- MuleSoft + Salesforce Financial Services Cloud: Core banking integration layer.

Banking Impact Metrics

Impact Area	Before	After	Business Impact
Duplicate Customer Records	10–14%	<2%	↑ Reliable identity for KYC/AML and product targeting
Onboarding Discrepancy Resolution Time	Several hours per case	<30 mins/case	↓ 75% operational effort
False Positives in Fraud Detection	Frequent	Reduced	↓ Manual review effort and fraud leakage
Cross-Sell Eligibility Errors	Common due to identity mismatches	Rare	↑ Targeting accuracy for product offers
Data Quality for AML Reporting	Inconsistent	Inconsistent	↑ Compliance and audit readiness
Customer Experience Friction	High due to ID mismatches	Minimal	↑ Satisfaction and retention



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